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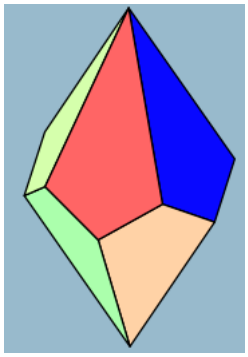
Pentagonal trapezohedron

In geometry, a **pentagonal trapezohedron** is the third in an infinite series of face-transitive polyhedra which are dual polyhedra to the antiprisms. It has ten faces (i.e., it is a decahedron) which are congruent kites.

It can be decomposed into two pentagonal pyramids and a pentagonal antiprism in the middle. It can also be decomposed into two pentagonal pyramids and a dodecahedron in the middle.

Pentagonal trapezohedron	
	
Type	trapezohedra
Symmetry group	D_{5d}, [2^+,10], (2*5), order 20
Rotation group	D_5, [2,5]$^+$, (225), order 10
Dual polyhedron	pentagonal antiprism
Properties	convex, face-transitive

10-sided dice

Ten ten-sided dice

The pentagonal trapezohedron was patented for use as a gaming die (i.e. "game apparatus") in 1906.[1] These dice are used for role-playing games that use percentile-based skills; however, a twenty-sided die can be labeled with the numbers 0-9 twice to use for percentages instead.

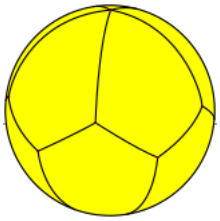
Subsequent patents on ten-sided dice have made minor refinements to the basic design by rounding or truncating the edges. This enables the die to tumble so that the outcome is less predictable. One such refinement became notorious at the 1980 Gen Con[2] when the incorrectly thought to cover ten-sided dice in general.

Ten-sided dice are commonly numbered from 0 to 9, as this a be rolled in order to easily obtain a percentile result. Where one die represents the 'tens represents 'units' therefore a result of 7 on the former and 0 on the latter would be combined to A result of double-zero is commonly interpreted as 100. Some ten-sided dice (often called 'Perc are sold in sets of two where one is numbered from 0 to 9 and the other from 00 to 90 in incre thus making it impossible to misinterpret which one is the tens and which the units die. Ten-sid also be marked 1 to 10 when a random number in this range is desirable.

Dice are small, throwable objects with marked sides that can rest in multiple positions. They are used for generating random values, commonly as part of tabletop games, including dice games, board games, role-playing ga

Spherical tiling

The pentagonal trapezohedron also exists as a spherical tiling, with 2 vertices on the poles, and alternating vertices equally spaced above and below the equator.



See also

Family of *n*-gonal trapezohedra [hide]

Trapezohedron name	Digonal trapezohedron (Tetrahedron)	Trigonal trapezohedron	Tetragonal trapezohedron	Pentagonal trapezohedron	Hexagonal trapezohedron	Heptagonal trapezohedron	Octagonal trapezohedron	Decagonal trapezohedron	Do tra
Polyhedron image									
Spherical tiling image									
Face configuration	V2.3.3.3	V3.3.3.3	V4.3.3.3	V5.3.3.3	V6.3.3.3	V7.3.3.3	V8.3.3.3	V10.3.3.3	

References

1. U.S. patent 809,293 (https://patents.google.com/patent/US809293)

2. "Greg Peterson about Gen Con 1980: *The big news of the year was that someone had 'invented' the ten-sided die*" (https://web.archive.org/web/20160814000600/http://planescape.outshine.com/official.planescape-torment.org/oldnews/greg_0507.html). Archived from the original (http://planescape.outshine.com/official.planescape-torment.org/oldnews/greg_0507.html) on 2016-08-14.

Sources

- Cundy, H. M.; Rollett, A. P. (1981). *Mathematical models* (3rd ed.). Tarquin. p. 117.

External links

- Generalized formula of uniform polyhedron (trapezohedron) having 2n congruent right kite faces (https://www.academia.edu/11805176/Mathematical_analysis_of_uniform_polyhedron_trapezohedron_having_2n_congruent_right_kite_faces_4n_edges_and_2n_2_vertices_lying_on_a_spherical_surface_generalized_formula_for_uniform_polyhedrons_with_congruent_right_kite_faces_) from Academia.edu
- Weisstein, Eric W. "Trapezohedron" (https://mathworld.wolfram.com/Trapezohedron.html). *MathWorld*.
- Virtual Reality Polyhedra (http://www.georgehart.com/virtual-polyhedra/vp.html) www.georgehart.com: The Encyclopedia of Polyhedra
  - VRML model (http://www.georgehart.com/virtual-polyhedra/vrml/pentagonal_trapezohedron.wrl)
  - Conway Notation for Polyhedra (http://www.georgehart.com/virtual-polyhedra/conway_notation.html) Try: "dA5"

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